

		$\frac{1}{r^3} \propto \frac{1}{T^2}$ (6)
9	A	Using $KE = \frac{GMm}{2R}$ and $TE = -\frac{GMm}{2R}$, both satellites will have the same TE if their KE is equal. Hence A is false. Since KE is same, mass of satellite is proportional to R. So, if X has a larger mass, it will have a larger radius of orbit. Option B is valid. Kepler's 3rd law would suggest that X has a larger period. Since angular velocity is inversely proportional to period, the former is smaller.
10	A	Work done in moving from X to Z